

how to get the bone surface in 3D slice?

I'm checking the most reliable way to do this in **3D Slicer**, since "bone surface" can mean either a quick display of the outer cortex or a clean exported 3D mesh.

Thought for 8s

If you mean **3D Slicer**, the usual way is:

To just see the bone surface

Use **Volume Rendering** on a CT scan. This is the fastest way to visualize the outer bone surface, and Slicer community guidance notes that volume rendering is often enough when you only need visualization, not a true mesh export. 3D Slicer +1

To make a real 3D bone surface model

Use **segmentation**, then show/export the closed surface:

1. Load your CT
2. Open **Segment Editor**
3. Click **Add** to create a new segment
4. Use **Threshold**
 - Adjust until bone is selected
 - Slicer's Segment Editor supports threshold-based segmentation for structures like bone. 3D Slicer +1
5. Clean the result if needed:
 - **Islands** to keep the main bone piece
 - **Smoothing** to reduce stair-step noise
 - Sometimes **Grow from seeds** works better if thresholding alone is poor. 3D Slicer Comm... +1
6. Click **Show 3D**
 - This displays the segmentation as a 3D surface. 3D Slicer Comm... +1
7. For export:
 - Go to **Segmentations** or use subject hierarchy export
 - Export the segment as a **model** such as STL/OBJ/PLY. Slicer documentation states that segmentations can be represented as **closed surface models** and

converted/exported between representations. [3D Slicer +1](#)

If you want only the outer surface

For cortical bone, community guidance suggests:

- **Threshold**
- **Smoothing**
- **Island removal**
- Fill holes if thresholding creates gaps

This is the common workflow for getting the outside surface without internal cavities dominating the model. [3D Slicer Comm... +2](#)

Practical tip

- **Higher threshold** can create holes.
- **Lower threshold** can create extra bulges/extrusions.

That tradeoff is specifically described by Slicer community experts. [3D Slicer Comm...](#)

Very short workflow

CT → Segment Editor → Threshold → Islands/Smoothing → Show 3D → Export model

If you want, I can give you:

- a **step-by-step example** for femur/skull in 3D Slicer, or
- the **best threshold starting range** in HU for bone CT.

  Sources

sure, please

I'm pulling together a concrete Slicer workflow with a practical HU starting range, so you can try it immediately on a CT volume.

Thought for 11s

Absolutely — here's a **practical 3D Slicer workflow** for getting a **bone surface** from a **CT**.

Example workflow in 3D Slicer

1. Load the CT scan

Import your DICOM or CT volume into Slicer.

2. Open Segment Editor

Create a new segment with **Add**. Segment Editor is the standard Slicer module for thresholding, painting, smoothing, and generating a 3D segment display. 3D Slicer

3. Use Threshold

For bone, a common **starting point** is roughly:

- **Lower threshold:** about **200 to 300 HU**
- **Upper threshold:** maximum intensity in the scan

Slicer community guidance and Slicer-based tutorials commonly use bone thresholds starting around **200 HU** or **300 HU**, depending on scan quality, body part, reconstruction kernel, and whether you want cancellous bone included. 3D Slicer Comm... +2

4. Click Apply

This creates a bone segment from the selected intensity range.

5. Clean the segment

Use these tools as needed:

- **Islands** → remove tiny disconnected pieces or keep the largest connected object
- **Smoothing** → reduce jagged staircase artifacts
- **Scissors / Erase / Paint** → manual cleanup if nearby dense structures were included

Slicer's Segment Editor includes these effects specifically for refining threshold-based segmentations. 3D Slicer

6. Click Show 3D

This generates the **closed surface** view of the segmented bone, which is usually what people mean by the "bone surface" in Slicer. Slicer's segmentation system supports both labelmap and closed-surface representations. 3D Slicer

7. Export the surface model if needed

If you need STL/OBJ/PLY for CAD or 3D printing, export the segmentation as a **model** from the Segmentations/Subject Hierarchy workflow. Slicer supports conversion of segmentations into closed-surface models for export. 3D Slicer

Good HU starting ranges for bone

These are **starting ranges**, not fixed rules:

Goal	Suggested lower threshold
Any bone, easy first pass	~200 HU
Cleaner cortical bone surface	~300 HU
Very dense cortical bone only	higher than 300 HU, adjusted case by case

That said, Slicer experts repeatedly note that **the correct threshold is not universal**. It changes with scanner calibration, reconstruction settings, anatomy, noise, and whether voxel values are true Hounsfield Units. 3D Slicer Comm... +2

Why one threshold does not always work

A bone threshold that works for one scan may fail on another because of:

- different CT acquisition and reconstruction settings
- noise and partial-volume effects
- whether the image values are correctly interpreted as **HU**
- desire to include trabecular bone versus only outer cortical bone

Slicer community discussions specifically warn that threshold values vary between series, and that segmentation quality can break down when scans are noisy or intensities are not mapped as expected. 3D Slicer Comm... +3

Best way to get only the outer surface

If your goal is the **outside bone shell**, not every internal cavity:

1. Start with **Threshold**
2. Use **Smoothing**
3. Remove unwanted floating parts with **Islands**
4. Manually patch holes if the cortex is broken
5. Then view in **3D**

This is the standard practical approach for outer bone surface extraction in Slicer-based workflows. Thresholds that are too high can create holes; thresholds that are too low can

create bulges or include non-bone structures. [3D Slicer Comm...](#) +2

Simple recommended starting recipe

For a normal CT bone surface attempt:

- **Threshold:** 300 HU to max
- **Smoothing:** light smoothing
- **Islands:** keep largest island if you want one main bone
- **Show 3D**

If that loses thin cortical regions, lower the threshold toward **200 HU**. If too much soft tissue/noise gets included, raise it gradually. This matches both Slicer community recommendations and published Slicer-based segmentation examples. [Amegroups](#) +2

Very short answer

Segment Editor → Add segment → Threshold bone (~200–300 HU to max) → Apply → Islands/Smoothing → Show 3D

I can also give you a **step-by-step skull example** or a **femur example** in Slicer.



sure, please

I'm putting this into a hands-on sequence you can follow click by click in Slicer, with one version for **skull** and one for **long bone/femur**.

Thought for 11s

Great — here are two **click-by-click examples** in 3D Slicer.

Example 1: Skull bone surface from CT

Goal

Create a 3D outer skull surface that you can view or export as STL/OBJ. The standard Slicer workflow is **Segment Editor → Threshold → cleanup → Show 3D**. [3D Slicer](#) +1

Steps

1. Load the CT

Open your CT volume in Slicer. Bone surface extraction works best on **CT**, because bone is usually separable by intensity thresholding much more reliably than on MRI. 3D Slicer +1

2. Open Segment Editor

Create a new segmentation and click **Add** to make a segment called `skull`. Segment Editor is the main Slicer tool for thresholding, smoothing, islands cleanup, and 3D display. 3D Slicer +1

3. Apply Threshold

Start with:

- **Lower:** about 200–300 HU
- **Upper:** maximum

For skulls, a lower threshold that is too high can create holes in thin regions like the orbital roof and skull base, while a lower threshold that is too low can include extra non-bone material. 3D Slicer Comm... +1

4. Click Apply

You should now see most of the skull selected. If the scan includes jaw, vertebrae, table, or other dense structures, they may come in too. That is normal in a first pass. 3D Slicer Comm... +1

5. Remove unwanted pieces

Use:

- **Islands** → **Keep largest island** if you only want the main skull
- or **Islands** → remove small disconnected parts
- **Scissors** for obvious unwanted regions

These are the common cleanup tools for threshold-based bone segmentation in Slicer. 3D Slicer +1

6. Fix holes

If your skull has many small holes:

- first try lowering the threshold a bit
- then use light **Smoothing**

- patch remaining problem areas with **Paint**

Slicer community guidance for skull/base segmentation notes that thin bone is especially vulnerable to holes when the threshold is too aggressive. [3D Slicer Comm... +2](#)

7. Click Show 3D

This generates the 3D closed surface from the segment so you can inspect the skull surface directly in 3D. [3D Slicer +1](#)

8. Export the skull surface

Export the segmentation as a **model** if you need STL/OBJ/PLY for CAD or 3D printing. Slicer's segmentation workflow supports generating and exporting closed-surface models.

[3D Slicer +1](#)

Good skull starting settings

A practical first attempt is:

- Threshold: **250–300 HU to max**
- Islands: **Keep largest island**
- Smoothing: **light only**

For skulls, avoid heavy smoothing if anatomical detail matters. Recent Slicer discussion also notes that smoothing/conversion settings can affect geometric accuracy.

[3D Slicer Comm... +1](#)

Example 2: Femur bone surface from CT

Goal

Extract one femur as a clean 3D bone model. Long bones are often easier than skulls, but touching anatomy can make separation harder. [3D Slicer Comm... +1](#)

Steps

1. Load the CT

Open the leg/hip CT in Slicer. For femur work, CT thresholding is the usual starting point.

[3D Slicer Comm... +1](#)

2. Open Segment Editor

Create a new segment called **Femur**. [3D Slicer](#)

3. Threshold the bone

Start around:

- **Lower:** 200–300 HU
- **Upper:** maximum

This should capture cortical bone and much of the cancellous region depending on scan quality and threshold choice. [3D Slicer Comm...](#) +1

4. Separate the femur from nearby bones

If pelvis, tibia, or table are connected, try:

- **Islands** if the femur is already disconnected
- **Scissors** to cut bridges
- **Grow from seeds** if threshold alone cannot separate touching bones well

Slicer examples and community guidance often use masking/seeded methods when thresholding alone cannot isolate the femur cleanly. [YouTube](#) +1

5. Fill cortical gaps

If the cortical shell has holes and you want only the outside surface:

- lower threshold slightly
- use light **Smoothing**
- patch gaps manually where necessary

This exact issue is common in femur/tibia threshold segmentation when exporting an outer surface model. [3D Slicer Comm...](#) +1

6. Show the femur in 3D

Click **Show 3D** and rotate the model to check for:

- holes in cortex
- attached extra structures
- jagged surface noise [3D Slicer](#)

7. Export

When it looks correct, export as STL/OBJ/PLY. [3D Slicer](#) +1

Good femur starting settings

A good first pass is:

- Threshold: **250 HU to max**
 - Islands: keep the femur island or remove small islands
 - Smoothing: light
 - Manual touch-up near joints or thin cortex if needed 3D Slicer Comm... +1
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Which example should you use?

Use the skull workflow if:

- you want cranium or facial bone surface
- you need a printable skull model
- thin bone holes are your main problem 3D Slicer Comm... +1

Use the femur workflow if:

- you want one long bone isolated
 - the main problem is separating one bone from nearby bones
 - you need a cleaner single-part orthopedic model 3D Slicer Comm... +1
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Fast troubleshooting

Too many holes

Lower the threshold a little and reduce smoothing aggressiveness. Thin bone regions are often lost when the threshold is too high. 3D Slicer Comm... +1

Too much junk included

Raise the threshold slightly, then use **Islands** and **Scissors**. 3D Slicer Comm... +1

Surface looks shifted or overly processed

Check segmentation conversion/smoothing settings; recent Slicer discussion notes some settings can affect surface accuracy. 3D Slicer Comm...

Scan is MRI, not CT

Bone extraction is much harder; CT is generally recommended for threshold-based bone segmentation. 3D Slicer Comm...

Best simple recipe

For most CT bone cases in Slicer:

1. Segment Editor
2. Add
3. Threshold: 200–300 HU to max
4. Apply
5. Islands
6. Light Smoothing
7. Show 3D
8. Export model 3D Slicer +2

I can give you next a **full skull tutorial with exact button names** or a **Slicer Python script** that automates the threshold-to-3D workflow.

